PROJECT REPORT OF INDUSTRY ORIENTED HANDS-ON EXPERIENCE (IOHE)

ON

ARTIFICIAL INTELLIGENCE BASED FRAMEWORK FOR ACADEMIC PERFORMANCE VISUALISATION

submitted in partial fulfilment of the requirements for the award of degree of

BACHELOR OF ENGINEERING

In

COMPUTER SCIENCE AND ENGINEERING

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DECLARATION

I hereby certify that the work which is being presented in the project report entitled **"ARTIFICIAL INTELLIGENCE BASED FRAMEWORK FOR ACADEMIC PERFORMANCE VISUALISATION"** in partial fulfilment of requirement for the award of the degree of Bachelor of Engineering (Computer Science and Engineering) submitted in the department of Computer Science and Engineering at Chitkara University Institute of Engineering and Technology, Chitkara University, Punjab, India, is an authentic record of my own work carried out under the supervision of Dr. Rajat Takkar. The matter presented in this project report has not been submitted in any other university/institute for the award of any degree.

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This is to certify that the above statement made by the candidate is correct to the best of my knowledge and belief.

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The project work and implementation were carried out using the Jupyter Notebook platform with the help of Python libraries such as NumPy, Pandas, Matplotlib, Seaborn, and Scikit-learn, which greatly facilitated data analysis, visualization, and model development.

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1. ABSTRACT

This project focuses on analyzing and predicting student academic performance using Machine Learning techniques. The system uses various academic parameters such as study hours, attendance, assignments, internal marks, and previous GPA to predict final scores, classify results (Pass/Fail), and determine student risk levels (Low, Medium, High).

A synthetic dataset is generated and analyzed using visualization tools to identify patterns and relationships between different academic factors. Machine Learning models such as Linear Regression and Decision Tree Classifier are implemented for prediction and classification tasks.

The project helps in identifying at-risk students early so that necessary actions can be taken to improve their performance. This system can be useful for educational institutions to enhance student success rates.

2. INTRODUCTION

In the modern education system, evaluating and improving student performance has become an essential task for educational institutions. With the increasing availability of academic data, it is now possible to analyze student behavior and performance using data-driven techniques. Traditional methods of assessment often fail to identify students who are at risk of poor performance at an early stage. Therefore, there is a need for intelligent systems that can predict outcomes and assist educators in decision-making.

This project focuses on **Student Performance Prediction and Risk Analysis using Machine Learning**. The main objective is to analyze various academic factors such as study hours, attendance, assignments, internal marks, and previous Grade Point Average (GPA), and use them to predict a student's final score. In addition, the system classifies students into categories such as Pass/Fail and determines their risk level (Low, Medium, High), which helps in identifying students who require additional academic support.

Machine Learning techniques provide an efficient way to discover patterns and relationships within data. In this project, algorithms such as Linear Regression and Decision Tree Classifier are used to perform prediction and classification tasks. These models are trained on a dataset and are capable of making accurate predictions based on input features.

The implementation of this system involves data generation, preprocessing, visualization, model training, and evaluation. Visualization tools are used to better understand how different factors influence student performance. The insights obtained from this analysis can help educators take proactive measures to improve student outcomes.

Overall, this project demonstrates how Machine Learning can be effectively applied in the field of education to enhance performance analysis, support decision-making, and ultimately improve the quality of learning.

3. METHODOLOGY

The methodology of the project includes the following steps:

3.1 Data Generation

A synthetic dataset is created using NumPy containing features like:

- Study Hours
- Attendance
- Assignments
- Internal Marks
- Previous GPA
- Final Score

3.2 Data Preprocessing

- Data is structured into a Pandas DataFrame
- Risk levels are assigned based on score ranges:
    - Low Risk
    - Medium Risk
    - High Risk

3.3 Data Visualization

Graphs are plotted using Matplotlib and Seaborn to analyze:

- Study hours vs performance
- Attendance impact
- Result distribution
- Risk level distribution

3.4 Model Building

- **Linear Regression** → Predict Final Score
- **Decision Tree Classifier** →
    - Classify Pass/Fail
    - Predict Risk Level

3.5 Model Evaluation

- Accuracy of classification model is calculated
- Predictions are tested with sample inputs

4. TOOLS AND TECHNOLOGIES

This project uses a combination of data analysis, machine learning, and visualization tools to build an efficient **Student Performance Prediction and Risk Analysis System**.

Programming Language

• **Python** is used as the primary programming language due to its simplicity and strong support for data analysis and machine learning.

Libraries and Frameworks

• **NumPy & Pandas** → used for data generation, handling, and preprocessing

• **Scikit-learn** → used for implementing machine learning models such as Linear Regression and Decision Tree

• **Matplotlib & Seaborn** → used for data visualization, including graphs and performance analysis

Machine Learning Techniques

• **Linear Regression** → used for predicting student final scores

• **Decision Tree Classifier** → used for classifying students into Pass/Fail categories and predicting risk levels

Other Technologies

• **Jupyter Notebook** → used as the development environment for coding, testing, and visualization

5. IMPLEMENTATION

The implementation of the project is divided into multiple steps:

5.1 Dataset Creation

A synthetic dataset is generated using random values to simulate real student data.

5.2 Feature Engineering

Features such as study hours, attendance, assignments, internal marks, and GPA are used as input variables.

5.3 Risk Level Classification

Students are categorized into:

- Low Risk (High Score)
- Medium Risk
- High Risk (Low Score)

5.4 Visualization

Different plots are created:

- Bar charts for performance analysis
- Pie charts for pass/fail distribution
- Risk level visualization

5.5 Model Training

- Linear Regression model predicts final score
- Decision Tree model predicts:
    - Pass/Fail
    - Risk Level

5.6 Prediction Functions

Custom functions are created:

- student_prediction() → Predict score
- predict_risk() → Predict risk level
- predict_risk_with_probability() → Risk with probability

6. MAJOR FINDINGS / RESULTS

The results of the project demonstrate that Machine Learning models can effectively analyze and predict student performance based on academic factors.

Key Results

• The system successfully predicts student final scores using Linear Regression
• Decision Tree Classifier accurately classifies students into Pass/Fail categories
• Risk levels (Low, Medium, High) are effectively identified based on performance

Observations

• Study hours and previous GPA have a strong positive impact on final scores
• Attendance plays a significant role in determining student performance
• Students with low study hours and attendance are more likely to fall into high-risk categories

Model Performance

• Linear Regression provides reliable score predictions
• Decision Tree Classifier performs well in classification tasks
• The model shows consistent and accurate results on test data

Visualization Insights

• Clear patterns are observed between study hours and performance
• Attendance and internal marks show a direct relationship with final scores
• Risk level distribution helps in identifying weak students easily

7. CONCLUSION AND FUTURE SCOPE

Conclusion

This project successfully demonstrates the application of Machine Learning techniques in analyzing and predicting student performance. By using academic factors such as study hours, attendance, assignments, internal marks, and previous GPA, the system is able to accurately predict final scores and classify students into Pass/Fail categories.

The Decision Tree Classifier effectively identifies student risk levels (Low, Medium, High), enabling early detection of students who may require additional academic support. Data visualization further enhances understanding by highlighting patterns and relationships between different factors affecting performance.

Overall, the model proves to be efficient, simple, and practical for educational use, helping institutions make better decisions to improve student outcomes.

Future Scope

- Use real-world datasets instead of synthetic data
- Improve accuracy using advanced models (Random Forest, XGBoost)
- Develop a web application using Flask or Django
- Integrate with college management systems
- Add real-time prediction dashboard

8. REFERENCES

- Rani, D. P., Venkatraman, S., & Bamrah, S. K. (2025). GradePulse - A Generative AI-Assisted Framework for Automated Student Performance Analytics.
- Shehab, J. M., Marni, N., Al Zitawi, D. U. D., & Kılıç, M. A. (2025). AI in Enhancing Student Academic Performance: A Bibliometric Analysis.
- Choi, W. C., Lam, C. T., & Mendes, A. J. (2024). Enhance Learning Performance Predictions with Explainable Machine Learning.
- Gull, H., Saqib, M., Iqbal, S. Z., & Saeed, S. (2020). Improving Learning Experience of Students by Early Prediction of Student Performance using Machine Learning.
- Nawahdah, M., Sawalha, H., Salameh, R., & Taha, M. (2025). Evaluating the Accuracy and Effectiveness of AI-Based Grading in Computer Science Education.
- Anoopkumar, M., & Rahman, A. M. J. M. Z. (2016). A Review on Data Mining Techniques and Factors Used in Educational Data Mining to Predict Student Performance.
- Wei, L. Y., & Nandy, T. (2025). Student Performance Prediction System Using Machine Learning.
- Bonde, S. N., & Kirange, D. K. (2018). Survey on Evaluation of Student Performance in Educational Data Mining.